

Citation under Pressure: What Deployment Constraints Do to Legal Citation Integrity, and What Deterministic Verification Can Repair

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Abstract. Sanctions over fabricated legal citations have made language-model drafting a problem for courts, and most research on the problem has produced detection benchmarks that score errors in finished briefs. This paper measures how the errors are produced. Seven models from seven vendors, from a 30B open-weight model to frontier systems, drafted argument sections for 48 U.S. Supreme Court matters under five prompting conditions: baseline, a citation quota, a restriction to pre-1970 authority, a sanctions warning, and all three combined. Every citation in the 1,680 drafts was checked deterministically, with no language-model judge in any primary measurement: existence against an 18-million-row reporter index, quotations against a verbatim checker validated on seeded faults, and pinpoint pages against official reporter pagination from public archives. Under combined pressure the smallest model’s citation existence fell from 0.924 to 0.773, and verbatim quotation accuracy fell for four models in the pre-registered run and for six of seven once three generations are pooled; Sonnet 5 was the exception, its accuracy numerically higher under the temporal, stakes, and combined conditions and, on the appellate task, higher under the sanctions warning after correction. In a second experiment the checker fed its findings back into revision. Sonnet 5 reached 0.942 verbatim accuracy; a per-quotation trace locates the rise. Of its 40 flagged quotations, 8 were corrected and most of the rest deleted; a flag that carries the closest passage of the cited opinion raises the corrections to 21 of 40, the one setting in which Sonnet 5 repaired more than it removed, while DeepSeek still removed 122 of 138. False feedback produced the same deletions aimed at correct quotations, four of seven models ended less accurate than they started, and a verifier right half the time removed twice the correct quotations that revision with no feedback did. In a separate task on 48 federal appellate matters, baseline existence was lower for six of seven models. We release the instruments, including a page-level pinpoint verifier from public data.

Keywords. legal citation, hallucination, large language models, verification, evaluation

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1. Introduction

Since 2023, courts have repeatedly sanctioned lawyers who filed briefs with citations a language model invented, beginning with *Mata v. Avianca* [1] and continuing in a docket a public tracker now counts in the thousands [2]. Most published work frames the problem as detection: given a brief that already contains errors, how many can a model or agent find [3,4]? Detection accuracy is the right measure for reviewing a filed brief; it does not tell a deployer which prompting conditions produce the errors, this paper’s question.

This paper asks two production-side questions. First, which prompting conditions cause citation failure, and for which models? Deployed prompts carry constraints of the kind Zhao et al. [5] studied for scholarly citations, where a publication-year window lowered citation existence even in capable models while a breadth demand and a non-disclosure instruction moved it little or raised it; we add a sanctions warning and know of no comparable study for law. Second, when a deterministic checker feeds its findings back into revision, does the model correct them, leave them, or change the draft so that the tests pass without the quotations being corrected?

Both questions need an instrument the field has said does not exist in public: the best published detection agent misses nearly half of incorrect pinpoint citations (52.8 percent recall), and its authors conclude that “no models can reliably detect wrong pincites” because page-level information lives behind commercial databases [3]. We built it from public sources: existence against a local index derived from CourtListener bulk data, verbatim quotation checking validated on seeded faults and audited on real drafts, and page-level pinpoint verification for U.S. Reports authority against the official pagination in the Caselaw Access Project’s volume archives [6].

Our primary setting is the closed-book use of general-purpose models, the setting of most sanctioned filings documented to date [2]; a grounded arm, in which each prompt carries ten retrieved and verified authorities, measures what retrieval changes. Drafts under the four non-baseline conditions are the pressured drafts (1,344 in the Supreme Court task, with three refusals), and the models with the highest baseline existence are the top of the range (Section 3).

2. Related Work

Legal citation reliability. Liu et al. [3] build a detection benchmark from real appellate briefs with injected errors and a taxonomy grounded in sanctioned filings; we adopt their error categories and measure the naturally occurring errors their limitations section asks for. LegalCiteBench [4] shows that models asked closed-book for a named case’s citation almost never produce it or abstain; our task asks the easier question, whether the cases a model chooses to cite exist. Dahl et al. [7] and Magesh et al. [8] measure hallucinated authority in chatbots and research tools.

Constraints and fabrication. Zhao et al. [5] is the closest design: five prompting regimes over scholarly claims, verified by fuzzy bibliographic matching that left 36 to 61 percent of citations unresolved. We transplant their factorial to law with deterministic verification (0.2 percent unresolved) and a formal statistical model.

Revision with execution feedback. The loop protocol follows execution-feedback work in code generation [9], including a no-feedback control that separates feedback

content from extra revision attempts; we add controls in which a quarter to all of the feedback is false.

AI and law. Blair-Stanek and Van Durme [10] found at ICAIL that frontier models flip the winning party across identical zero-temperature runs, which Section 8 bounds with repeated generations. Evaluation in this community pairs automated metrics with the checks lawyers perform [11,12]; citation integrity is one with deterministic ground truth. Garneau et al. [13] found retrieving the paragraph a proposition rests on far harder than retrieving the case; our verifier meets the same gap from the other side, checking a generated pincite against the page text. Zhang et al. [14] show models miss overruled precedent, adjacent to the wrong-authority errors we measure; ConsRAG [15] constrains retrieval, complementary to our loop’s feedback; Trozze et al. [16] and Harasta et al. [17] study how jurors and lawyers perceive model-written documents, a perception machine verification does not depend on. Evaluations of generated legal argument [18,19] do not check the authority it cites, the layer this paper adds; citation-worthiness and citation-network work [20,21,22,23] studies the network our checker queries.

3. Tasks, Conditions, and Instruments

Terms. In 347 U.S. 483, 495, the volume, reporter, and first page identify the case and the second page is the *pincite*, for a quotation the page where the quoted language appears. *Citation existence* is the share of adjudicable citations that resolve to a real case; *strict quotation accuracy* the share of scored quotations that reproduce the opinion verbatim.

Matters. The matters are forty-eight Supreme Court cases from 1990 to 2020 drawn from the Supreme Court Database [24]. Each packet holds the question presented, party names, a factual summary, and a lower-court opinion excerpt, the Supreme Court’s decision excluded and outcome leakage screened. The model argues an assigned side (petitioner for even-numbered matters, respondent for odd) in an argument section of under 900 words; nothing predicts outcomes, so the memorization confound documented for Supreme Court case classification [25] is out of scope. A second task uses forty-eight published F.3d decisions from the Caselaw Access Project archive (1995 to 2019, four from each of the First through Eleventh and Federal Circuits, no cross-appeals). The packet is the opinion’s own statement of the case and facts, cut where the analysis begins and scrubbed of outcome sentences, and the model argues for the named appellant under all five conditions.

Conditions. The five conditions are baseline; quota (“cite at least eight distinct case authorities”); temporal (“every authority you cite MUST have been decided before 1970”), a clause no user writes that stands in for the binding-authority and jurisdiction limits practitioners do impose, and which, unlike Zhao et al.’s window on recent work, moves citations toward well-memorized opinions; stakes (“this section will be filed with the court; citing authority that does not exist, or misquoting authority, risks sanctions under Rule 11”); and combined, which concatenates the three clauses. Each condition has one fixed template in the primary run, at temperature 0. The stakes clause names Rule 11, the rule invoked in the sanctioned-filing cases [1]; it governs district-court civil actions [26], so for a Supreme Court filing the clause is a stakes signal only. Each clause is a single sentence so that its effect can be isolated; a second template rephrasing every condition was run on all 48 matters and seven models. A *grounded* arm reran baseline

and combined with ten retrieved authorities in the prompt: the U.S. Reports opinions most similar to the question and facts (TF-IDF over the first 2,500 words of the 25,250 opinions in our Caselaw Access Project cache, decided before argument and before 1970 for the combined condition, any opinion naming either party excluded), each with citation, year, and its best-matching 250-word passage, stated to be real, quotation restricted to the excerpts, other authority permitted.

Models. The models are Qwen3-30B-A3B, Mistral Small, Llama-4 Maverick, DeepSeek V4 Flash, Grok 4.3, GPT-5.4-mini, and Claude Sonnet 5, one per vendor, spanning open-weight 30B to frontier; tables are ordered by baseline strict quotation accuracy, and Sonnet 5, GPT-5.4-mini, and Grok 4.3, with the highest baseline existence, are called the top of the range. The seven were run through OpenRouter in August 2026 with completion limits of 1,400 to 12,000 tokens; the identifiers are qwen3-30b-a3b-instruct-2507, mistral-small-2603, llama-4-maverick, deepseek-v4-flash, grok-4.3, gpt-5.4-mini, and claude-sonnet-5. An eighth model, GLM-4.7-flash, produced no output on 33 of 240 drafts (28 combined), exhausting a 30,000-token budget in reasoning, a third failure mode beside fabrication and refusal; it is excluded from the tables.

Adjudication. Citations are extracted with eyecite [27] and given one of three labels: exists, not found, or unverifiable (vendor citations and coverage gaps), so that index coverage is never counted as fabrication; 0.2 percent of the 14,040 citation records in the full run are unverifiable. Citations resolve against an 18-million-row table; quotations are located by balanced quotation marks and attributed, in order, to the citation whose parenthetical holds them, the case named in their own sentence, the citation that follows them in that sentence (not one introduced by *see* or *cf.*), the citation an *Id.* points to, or the last citation in the paragraph; a quotation introduced by a statute, rule, the record, or a lower court, or with no citation in its paragraph, is left unpaired (37 percent of the 10,846 quotations in the full run), since the checker has nothing to test it against. Quotations are checked verbatim against the cited opinion’s text from public sources. *Exists* means the volume, reporter, and page resolve to an indexed opinion; the case name is compared separately, and 205 of 11,769 resolving citations (1.7 percent) carry a name the index does not match, a ceiling that includes abbreviated names (Sonnet 5 6 of 1,787). A quotation is *near miss* when every fragment reaches token coverage of at least 0.85 without matching exactly, and *inaccurate* otherwise. The strict rate for a draft d is $S_d = n_{\text{acc}} / (n_{\text{acc}} + n_{\text{near}} + n_{\text{inacc}})$, and a lenient rate counting near misses as correct is reported alongside it. Matching is case and punctuation insensitive, splits on ellipses, and reads a bracketed alteration as its contents. The quotation checker scores 240 of 240 on twenty synthetic drafts, each planting twelve items, six of them attribution traps (a rival citation after a signal phrase, a statute, an *Id.*, a string-cite parenthetical, paragraphs with and without a citation), and the existence and pinpoint checks scored 100 of 100 on 20 seeded opinions (real, fabricated, and vendor citations; a sentence pinned to its own page and to one three or more away); Section 6 audits it on real drafts. Pinpoint pages are verified against the official reporter pagination in the Caselaw Access Project’s HTML volumes, which mark every page break; a quotation counts as on the cited page when its longest fragment lies within that page’s text.

Human reference. The same instrument scored 482 pre-ChatGPT appellate brief excerpts from Liu et al. [3] (no injected error; OCR-converted, so a floor); human lawyers reach 0.493 strict and 0.764 lenient quotation accuracy (416 scored quotations) and 0.931

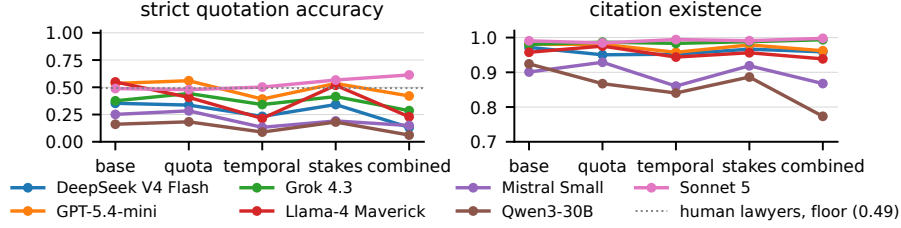


Figure 1. Strict quotation accuracy (left) and citation existence (right) by condition, one line per model; the dotted line is the human-lawyer floor under the same instrument (Section 3); the right axis starts at 0.7.

citation existence overall, 0.976 on U.S. Reports and published federal reporters (24 of the 61 not found are Federal Appendix citations, a reporter the index does not hold), a reference for the severity of the verbatim standard only, since human briefs differ in citation mix. Of the 211 human strict failures, 84 carry a bracketed alteration or ellipsis, 60 are near misses without one, and 67 are inaccurate without one; reading every bracketed segment as an omission lifts the human rate to 0.594 and moves model cells by at most 9 points, with the same 8 contrasts surviving correction (Section 4).

Inference. The primary analysis is a citation-level logistic generalized estimating equation per model, $\text{logit}P(y_{imk} = 1) = \beta_0 + \beta_k$ for citation i in matter m under condition k , with exchangeable working correlation within matters [28], so that the 48 matters are the independent units. Each model has 8 contrasts (four conditions, two outcomes); we report Holm-corrected p values within model and call an effect significant only when it survives correction; counts of models (“three of seven”) are descriptive. Odds ratios, intervals, and p values for all 56 contrasts are released; a matter-level cluster bootstrap agreed in sign with every corrected result. Three robustness fits follow the same model: one pools two further generations of the baseline and combined drafts (Section 8) with a run effect (Holm over 14 tests), the others pool both templates or both tasks with a template or task effect, corrected within model. For the revision experiment, pre-registered descriptively, we add matter-paired comparisons of final strict accuracy between arms (Wilcoxon, bootstrap interval). Hypotheses and decision rules were frozen before the full run; every later decision, including two withdrawn findings, is in a dated ledger released with the code.

4. Experiment 1: Pressure

Table 1 and Figure 1 report the rates.

Citation existence. In the single run, only the 30B model’s existence moves significantly under pressure: from 0.924 at baseline to 0.773 under the combined condition, with the quota, temporal, and combined contrasts all significant after correction (combined OR 0.30, 95 percent CI 0.18 to 0.51, corrected $p < 0.001$). Mistral Small falls under the temporal and combined conditions without reaching significance, and the five other models stay between 0.93 and 1.00 in every condition.

Quotation accuracy. The temporal restriction damages quoting most; after correction, strict quotation accuracy falls under temporal and combined pressure for Llama-4 (0.549 to 0.230), under the combined condition for Qwen3-30B (0.161 to 0.061) and

Table 1. Baseline versus the combined condition; quotation columns give the strict rate (lenient in parentheses), existence columns the rate [citations not found / adjudicable]; an asterisk marks a contrast significant after Holm correction within model.

Model	Quotes, base	Quotes, comb.	Exist, base	Exist, comb.	Viol., temporal
Qwen3-30B	0.161 (0.609)	0.061 (0.320)*	0.924 [28/370]	0.773 [90/397]*	20%
Mistral Small	0.251 (0.626)	0.149 (0.444)	0.901 [37/373]	0.868 [69/521]	35%
DeepSeek V4 Flash	0.353 (0.726)	0.132 (0.491)*	0.972 [10/354]	0.959 [20/488]	11%
Grok 4.3	0.376 (0.713)	0.286 (0.647)	0.981 [5/257]	0.994 [3/479]	0%
Sonnet 5	0.488 (0.800)	0.613 (0.825)	0.991 [3/318]	0.998 [1/450]	0%
Llama-4 Maverick	0.549 (0.745)	0.230 (0.574)*	0.958 [9/213]	0.939 [31/506]	10%
GPT-5.4-mini	0.535 (0.795)	0.420 (0.693)	0.981 [8/417]	0.962 [27/713]	13%

DeepSeek (0.353 to 0.132), and under the temporal clause for Mistral Small; GPT-5.4-mini and Grok 4.3 fall in the same direction without surviving correction in the single run, and Sonnet 5 does not fall (below). Llama-4 shows the largest drop. At baseline it quotes above the human floor, 0.549 against 0.493, and under the temporal restriction its accuracy drops to 0.214 (OR 0.23, CI 0.09 to 0.61, corrected $p=0.022$); refitting every contrast on the lenient outcome, 8 survive correction, the combined fall among them for Qwen3-30B, DeepSeek, and Mistral Small (Llama-4’s reaches corrected $p=0.583$). Nor is the fall an artifact of scanned text for old opinions; at baseline, strict accuracy for quotations from opinions decided before 1970 is 0.392 (189 quotations) against 0.354 for later ones (1,075). With three generations pooled, the combined-condition fall survives correction for 6 of the seven models (OR 0.23 to 0.57), Sonnet 5 excepted, and Mistral Small’s existence fall also survives (OR 0.62, corrected $p=0.004$).

Robustness to phrasing. Under the second template, 9 contrasts survive correction, among them 5 of the 8 that survived under the first (Qwen3-30B’s quota existence fall and Llama-4’s two quotation falls did not); pooling both templates, all 8 survive and the combined-condition fall in quotation accuracy survives for 5 of seven models (OR 0.29 to 0.51), Sonnet 5 again the exception (pooled OR 1.28, corrected $p=0.904$). Cell rates differ between templates by up to 13 points of quotation accuracy and 19 of existence.

Second task: federal appellate briefs. Outside the Supreme Court, fabrication at a neutral prompt is more common. Baseline citation existence on the 48 appellate matters is 0.645 for Qwen3-30B, 0.725 for Mistral Small, between 0.907 and 0.959 for the four models in between, and 0.993 for Sonnet 5, lower than on the Supreme Court matters for 6 of the seven models with Sonnet 5 alone unchanged. The failures concentrate in the federal reporters: 22 percent of citations to F.2d, F.3d, and F. Supp. do not resolve, against 4 percent for the U.S. Reports; resolving the 1,766 citations in the 48 deciding opinions themselves, real by construction, the index misses 3 of 759 to those reporters (0.4 percent) and 1 of 227 to the U.S. Reports, so the shortfall is not coverage. Two pressure contrasts survive correction within the task: Llama-4’s existence fall under the temporal clause (0.945 to 0.857, OR 0.35, corrected $p=0.032$) and Sonnet 5’s quotation rise under the stakes clause alone (0.436 to 0.597, OR 1.88, $p=0.019$). Pooling both tasks with a task effect, six of the 8 Supreme Court survivors remain (Qwen3-30B’s quota and temporal existence falls drop out), joined by Llama-4’s existence falls under temporal and combined, DeepSeek’s temporal and Mistral Small’s combined quotation falls, and Sonnet 5’s rise under stakes (OR 1.55, corrected $p=0.022$).

Grounded drafting. With ten verified authorities in the prompt, the models drew 51 to 83 percent of their citations from the list; citations not found fell from 341 in 5,856 closed-book to 74 in 5,605, and Qwen3-30B’s combined-condition existence went from 0.773 to 0.995 (OR 53.8, corrected $p < 0.001$). Quotation accuracy rose in every combined cell and 7 of seven baseline cells (Qwen3-30B combined 0.061 to 0.407; Sonnet 5 baseline 0.488 to 0.656), yet with the source passages in the prompt no cell exceeded 0.718, and 62 percent of Sonnet 5’s 74 remaining inaccurate quotations are paraphrases of the cited case in quotation marks.

The strongest model does not degrade. Sonnet 5’s strict quotation accuracy is numerically higher under the temporal, stakes, and combined conditions and highest under the combined one (0.488 to 0.613; OR 1.65, CI 1.04 to 2.60, uncorrected $p = 0.0315$, corrected $p = 0.252$), the one pre-registered contrast in that direction; the stakes condition alone moves nothing after correction on these matters, and the temporal restriction shifts the cited cases toward older, better-memorized opinions (median decision year 1943 against 1986). With three generations pooled Sonnet 5’s existence rise survives the logistic fit (corrected $p = 0.006$), but it rests on 7 of 975 citations not found at baseline against 5 of 1,324 under combined, twelve events on which an exact test gives $p = 0.38$; the quotation rise reaches corrected $p = 0.454$ there. In the appellate task the stakes clause alone raises it after correction without moving the cited cases (median decision year 1991 at baseline and under stakes, 1946 under the temporal clause), which memorization does not cover, and the rise survives pooling both tasks; we report the direction as consistent and its cause as open.

Compliance and refusals. Violations of the pre-1970 rule under the temporal condition alone (Table 1) run from 35 percent for Mistral Small to 0 of 353 for Sonnet 5 and do not track the other outcomes in rank; the 1,344 pressured drafts held three refusals (two DeepSeek, one GPT-5.4-mini).

5. Experiment 2: The Checker inside the Revision Loop

Combined-condition drafts entered a revision loop: each round the checker returned one line per failure (a citation that does not resolve; a quotation not verbatim in the cited opinion), the model revised, and the loop ended on a clean draft or after three revisions. A *false feedback* control gave the same number of identically formatted lines aimed at correct items; *mixed* controls replaced each line with a false one at random so that precision was about 0.25, 0.5, or 0.75; and a *no-feedback* control gave the same rounds with a generic instruction to revise. The loop used the first 24 matters in pre-registered order; loop rates are means over drafts with at least one scored quotation (a draft left with none drops out; 60 of the 168 true-arm episodes ended so, 19 of Grok 4.3’s 24, so its final rate rests on 5 drafts), so round-0 figures differ from the pooled rates in Table 1. Near misses are 195 of the 507 round-0 flags, so some true feedback targeted a bracketed alteration. Every scored round-0 quotation was traced to the final draft as *corrected* (an accurate quotation on the same citation now stands in its place, edited or new), *dequoted* (words kept, marks dropped), *deleted*, or left in place.

Satisfying the checker by deletion. Under true feedback the final strict rate ranges from 0.096 for Qwen3-30B to 0.942 for Sonnet 5 (Table 2). Sonnet 5 had 40 flagged quotations at round 0: 3 were corrected in place and 5 replaced by a new accurate quo-

Table 2. Revision under true checker feedback, 24 episodes per model. Round-0 quotations, flagged (inaccurate or near miss) and accurate, traced to the final draft; corr.: corrected, an accurate quotation on the same citation now stands in its place, edited or new; left: still present and flagged; removed: deleted, dequoted, or made inaccurate; added: quotations new in the final draft, accurate/flagged; strict accuracy at round 0 and final as the draft mean (pooled over quotations); Clean: episodes ending on a passing draft.

Model	Flagged at round 0					Accurate at round 0					Strict, r0 → final (pooled)	Clean
	n	corr.	dequoted	deleted	left	n	removed	added				
Qwen3-30B	138	0	9	22	107	12	0	0/33	0.089 → 0.096 (0.079)	0		
Mistral	97	4	14	28	51	12	3	1/22	0.224 → 0.279 (0.161)	9		
DeepSeek	138	1	46	75	16	27	14	2/13	0.278 → 0.623 (0.356)	21		
Grok 4.3	37	0	21	15	1	13	6	0/0	0.234 → 0.800 (0.875)	24		
Sonnet 5	40	8	4	26	2	64	21	8/2	0.688 → 0.942 (0.937)	23		
Llama-4	33	0	3	21	9	11	6	1/2	0.273 → 0.477 (0.353)	21		
GPT-5.4-mini	24	1	1	16	6	29	6	1/1	0.505 → 0.698 (0.781)	22		

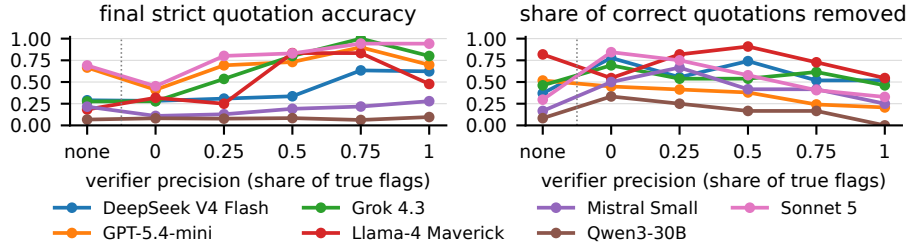


Figure 2. Verifier precision against what revision does. Left: final strict quotation accuracy. Right: share of accurate round-0 quotations that revision deleted, dequoted, or made inaccurate. Precision is the share of true feedback lines; the leftmost point is revision with no feedback.

tation from the same case, 4 kept their words and lost their quotation marks, and 26 were deleted; 21 of its 64 accurate quotations went too. Citation existence rose by the same route: of 113 citations unresolved at round 0 across the seven models, 86 were gone from the final draft and 27 remained, with 33 new resolving citations, 17 of them Mistral Small’s. Rerunning the loop from the grounded combined drafts of Sonnet 5 and DeepSeek, with the retrieved excerpts kept in the prompt, left repair at 3 of 29 flags and 5 of 55. A *passage* arm then put the evidence in the flag itself: each quotation line carried the closest passage of the cited opinion (the window search of Section 4). Repair more than doubled, to 21 of 40 for Sonnet 5 (5 in place, 16 replaced by the supplied passage) against 16 deleted or dequoted, the one setting in which it repaired more than it removed, and to 14 of 138 for DeepSeek against 1, which still removed 122; final strict rates were 0.920 and 0.909 over more surviving quotations (92 against 63 for Sonnet 5). A loop should hand the model the passage; below the top of the range it deletes most of what is flagged even then.

Controls: feedback content and verifier precision. Without feedback (Figure 2), Sonnet 5’s strict rate still rose (0.688 to 0.691), and true feedback added 0.23 over revision alone (paired over matters; interval 0.09 to 0.36, $p=0.010$), as for DeepSeek (0.25); Llama-4’s (0.20) and GPT-5.4-mini’s (0.01) intervals include zero. Sonnet 5’s correct quotations removed rise as precision falls, 21, 26, 37, 48, and 54 of 64 at precisions of 1,

0.75, 0.5, 0.25, and 0, against 19 with no feedback; at one half the final rate was 0.830 against 0.691 with none. The mixed arms lower recall with precision, so the left panel mixes errors left standing with correct quotations removed; the right panel isolates the latter. Under the lenient verdict (0.971 with true feedback, 0.855 with all-false, 0.884 with none) only all-false feedback leaves Sonnet 5 below revision alone. Under all-false feedback four of seven models ended below their starting rate and flagged counts fell in every model, so what changed was the deletion of correct quotations.

6. Residual Errors at the Top of the Range

Paraphrase inside quotation marks. Decomposing every inaccurate quotation against the cited opinion's text, 68 percent of Sonnet 5's (175 of 258) are recognizable paraphrases of the correct case presented as verbatim quotation, 56 match nothing substantial in the cited opinion, and 27 cite authority outside the U.S. Reports, which the decomposition does not cover.

Real language, wrong case. Searching every inaccurate quotation against the 62,049 U.S. Reports opinions in the Caselaw Access Project cache locates 330 quotations across the seven models (75 for Sonnet 5, 43 for Qwen3-30B) whose language appears verbatim in a different opinion than the one cited; in 140 of these the true source is also cited in the same draft, so the count is a ceiling; quotations found nowhere in it (134 for Sonnet 5) include the paraphrases above and quotations from other courts, and 44 of Sonnet 5's matched only stock phrases and are excluded. Two audits bound the checker's own error among inaccurate verdicts. The same search finds the language present in the attributed case for 5 of Sonnet 5's 258 and 2 of Qwen3-30B's; and a hand reading of forty inaccurate verdicts drawn at random across models found 8 on the checker's side (a policy name or statutory phrase in quotation marks, a lower court's words, a quotation bound to a case other than the one named), 30 the model's, and two open. The rule reached this precision through the audit; its first version, which let a later-named case take a quotation and bound statutory quotations to the nearest case, put about half of the verdicts on the checker's side.

Right case, wrong page. Of the 9,129 pinpoints in the Supreme Court drafts, 73 percent were checked against the cited case's page span and 1,006 carry a quotation on the same citation; for those the quotation lies on the cited page 83 percent of the time for Sonnet 5, 50 to 67 percent for the other six, and 88 percent for the human briefs (41 quotations); otherwise it lies on another page of the cited case, adjacent or not.

7. Discussion

Where the checker belongs. The revision experiment argues for the checker at the gate, returning failures to a person, and against it in the loop, where the model resolves them, and it says what a loop should carry if one is built, the passage in place of the bare flag; removing an unverifiable quotation is a defensible response to a flag, and three facts keep it from being repair. The quotation is deleted or survives as paraphrase the checker cannot test; false flags get the same response, so correct quotations go too (Figure 2); and a loop-pattern tool reports a clean draft, so the attorney never sees the list.

Verifier precision is a deployment requirement. Our checker’s verifiable error among Sonnet 5’s inaccurate verdicts is 1.9 percent by provenance and about 20 percent by the hand reading (Section 6); Liu et al. [3] report weaker detectors flagging correct citations by reading coverage gaps as fabrication. Figure 2 shows what such flags do: at precision one half Sonnet 5 lost 37 of its 64 correct quotations, against 21 under a precise verifier and 19 with no feedback. A pipeline that feeds a probabilistic detector’s flags to a model should report the detector’s precision beside its recall, and a deployer should enforce the lenient standard, which admits bracketed alterations and puts the human reference at 0.764.

What evaluations should include. A single fabrication rate at a neutral prompt misses what is reported here: existence under constraints of the kind Zhao et al. [5] document and on other courts, quotation accuracy, which existence benchmarks omit, and the residual classes at the top of the range, invisible to existence checks and surviving retrieval. The three checks can also be run by a court on any filing, and cover the nonexistent-authority and misquotation failures of the sanctions cases [1,2].

8. Limitations

Both tasks are United States federal case law, the Supreme Court task under two prompt templates and the appellate task under one; the stakes clause names a sanctions rule that does not govern the forum; and the checks stop at existence, wording, and page: relevance has no deterministic test, and three language-model panels calibrated on expert labels [3] reached 0.500 to 0.700 against a bar of 0.75, so no such verdict was used [29,30]. The controls, the grounded arm and loop, the second task, and the per-quotation trace were added after the pre-registered analysis, their tests chosen after the data were seen, and are reported separately from the pre-registered fit. Three generations of the baseline and combined cells (672 drafts per run) bound run-to-run variation [10]: no draft repeated (22 of Mistral Small’s excepted), cell standard deviations were at most 0.050 (existence) and 0.056 (strict), and the combined-versus-baseline quotation contrast kept its sign in every run for all 7 models; we rely on the contrasts that survive correction across runs and templates. Inference is per model at 48 matters, and the seven models are one per vendor, confounding capability, family, and recipe.

9. Conclusion

Under a citation quota, a date restriction, and a sanctions warning, citation integrity degrades from the smallest model upward, from nonexistent citations at 30B through misquotation in the middle to paraphrase in quotation marks, misattribution, and wrong pages at the top. Retrieval removes most existence failures and leaves paraphrase. Returned to the model, its findings raise existence chiefly by removal, flagged quotations are removed far more often than corrected, repair overtaking removal only at the top of the range with the passage in the flag, and false flags remove correct ones; verification belongs at the gate, and verifier precision in the evaluation of any system that acts on it. Instruments, drafts, records, and ledger are at <https://github.com/hardy30894/citation-under-pressure>.

References

- [1] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023); 2023. Opinion and order on sanctions, No. 1:22-cv-01461 (PKC), June 22, 2023.
- [2] Charlotin D. AI Hallucination Cases Database; 2025. Accessed 2 September 2026; 2,008 cases listed. <https://www.damiencharlotin.com/hallucinations/>.
- [3] Liu P, Stambach D, Henderson P. Who Checks the Citations? Benchmarking Legal Hallucination Detection; 2026. Accepted at the Conference on Language Modeling (COLM) 2026. Available from: <https://arxiv.org/abs/2606.21155>. doi:10.48550/arXiv.2606.21155.
- [4] Chen S, Yin H, Zhou S. LegalCiteBench: Evaluating Citation Reliability in Legal Language Models; 2026. Available from: <https://arxiv.org/abs/2605.10186>. doi:10.48550/arXiv.2605.10186.
- [5] Zhao C, Tang Y, Qian Y. Do Deployment Constraints Make LLMs Hallucinate Citations? An Empirical Study across Four Models and Five Prompting Regimes; 2026. Available from: <https://arxiv.org/abs/2603.07287>. doi:10.48550/arXiv.2603.07287.
- [6] Harvard Law School Library Innovation Lab. Caselaw Access Project; 2024. Digitized collection of official, book-published U.S. case law; accessed September 2026. <https://case.law/>.
- [7] Dahl M, Magesh V, Suzgun M, Ho DE. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis*. 2024;16(1):64-93. doi:10.1093/jla/laae003.
- [8] Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. *Journal of Empirical Legal Studies*. 2025;22(2):216-42. doi:10.1111/jels.12413.
- [9] Gehring J, Zheng K, Copet J, Mella V, Cohen T, Synnaeve G. RLEF: Grounding Code LLMs in Execution Feedback with Reinforcement Learning. In: *Proceedings of the 42nd International Conference on Machine Learning*. vol. 267 of *Proceedings of Machine Learning Research*. PMLR; 2025. p. 19034-55. Available from: <https://proceedings.mlr.press/v267/gehring25a.html>.
- [10] Blair-Stanek A, Van Durme B. LLMs Provide Unstable Answers to Legal Questions. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. Chicago, IL, USA: ACM; 2025. p. 425-9. doi:10.1145/3769126.3769245.
- [11] Gultekin S, Rossi Reich M, Galloni F, Lagioia F, Consiglio E, Sartor G, et al. A Two-Dimensional Evaluation Framework for Factual and Reasoning Assessment of LLMs in Legal Question Answering. In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 86-97. doi:10.3233/FAIA251579.
- [12] Sanchi M, Novotná T. Towards Hybrid Evaluation Methodologies for Large Language Models in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 389-92. doi:10.3233/FAIA241279.
- [13] Garneau N, Olsen HP, Corduant A, Tarissan F. Combining Network and Text to Provide Legal Pincites. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 282-7. doi:10.3233/FAIA241256.
- [14] Zhang L, Savelka J, Ashley KD. Do LLMs Truly “Understand” when a Precedent Is Overruled? In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 239-50. doi:10.3233/FAIA251592.
- [15] Nguyen HT, Satoh K. ConsRAG: Minimize LLM Hallucinations in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 327-32. doi:10.3233/FAIA241263.
- [16] Trozze A, Davies T, Kleinberg B. Large language models in cryptocurrency securities cases: can a GPT model meaningfully assist lawyers? *Artificial Intelligence and Law*. 2025;33(3):691-737. doi:10.1007/s10506-024-09399-6.
- [17] Harasta J, Novotná T, Savelka J. It cannot be right if it was written by AI: on lawyers’ preferences of documents perceived as authored by an LLM vs a human. *Artificial Intelligence and Law*. 2026;34(1):153-90. doi:10.1007/s10506-024-09422-w.
- [18] Gray MA, Savelka J, Oliver WM, Ashley KD. Generating Legal Arguments with Automatically Identified Factor Magnitudes. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 199-208. doi:10.1145/3769126.3769212.
- [19] Pires R, Malaquias Junior R, Nogueira R. Automatic Legal Writing Evaluation of LLMs. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 420-4. doi:10.1145/3769126.3769227.

- [20] Khatri M, Sheik R, Wadhwa P, Satija G, Kumar Y, Shah RR, et al. CiteCaseLAW: Citation Worthiness Detection in Caselaw for Legal Assistive Writing. In: Legal Knowledge and Information Systems: JURIX 2023, vol. 379 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2023. p. 257-62. doi:10.3233/FAIA230971.
- [21] Sadeghian A, Sundaram L, Wang DZ, Hamilton WF, Branting K, Pfeifer C. Automatic semantic edge labeling over legal citation graphs. Artificial Intelligence and Law. 2018;26(2):127-44. doi:10.1007/s10506-018-9217-1.
- [22] Dadgostari F, Guim M, Beling PA, Livermore MA, Rockmore DN. Modeling law search as prediction. Artificial Intelligence and Law. 2021;29(1):3-34. doi:10.1007/s10506-020-09261-5.
- [23] Mumford J, Florimonte F, Atkinson K, Dzehtsiarou K. Context-Aware Citation Networks: A Human-AI Dataset, Analysis, and Tool. In: Legal Knowledge and Information Systems: JURIX 2025, vol. 416 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2025. p. 145-55. doi:10.3233/FAIA251584.
- [24] Spaeth HJ, Epstein L, Nelson MJ, Martin AD, Segal JA, Ruger TJ, et al. Supreme Court Database, Version 2025 Release 01; 2025. Modern database, terms 1946 to 2024. <https://scdb.la.psu.edu/>.
- [25] Ortega JE, Joshi DD, Borkowski MP. Large-Language Memorization During the Classification of United States Supreme Court Cases; 2025. Available from: <https://arxiv.org/abs/2512.13654>. doi:10.48550/arXiv.2512.13654.
- [26] Federal Rules of Civil Procedure, Rule 1 (Scope and Purpose); 2024. "These rules govern the procedure in all civil actions and proceedings in the United States district courts". https://www.law.cornell.edu/rules/frcp/rule_1.
- [27] Cushman J, Dahl M, Lissner M. eyecite: A tool for parsing legal citations. Journal of Open Source Software. 2021;6(66):3617. Available from: <https://doi.org/10.21105/joss.03617>. doi:10.21105/joss.03617.
- [28] Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. Biometrika. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13.
- [29] Norman JD, Rivera MU, Hughes DA. Reliability without Validity: A Systematic, Large-Scale Evaluation of LLM-as-a-Judge Models Across Agreement, Consistency, and Bias; 2026. Available from: <https://arxiv.org/abs/2606.19544>. doi:10.48550/arXiv.2606.19544.
- [30] Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena; 2023. NeurIPS 2023 Datasets and Benchmarks Track. Available from: <https://arxiv.org/abs/2306.05685>. doi:10.48550/arXiv.2306.05685.